

# HMM POS TAGGER

## Team Member:

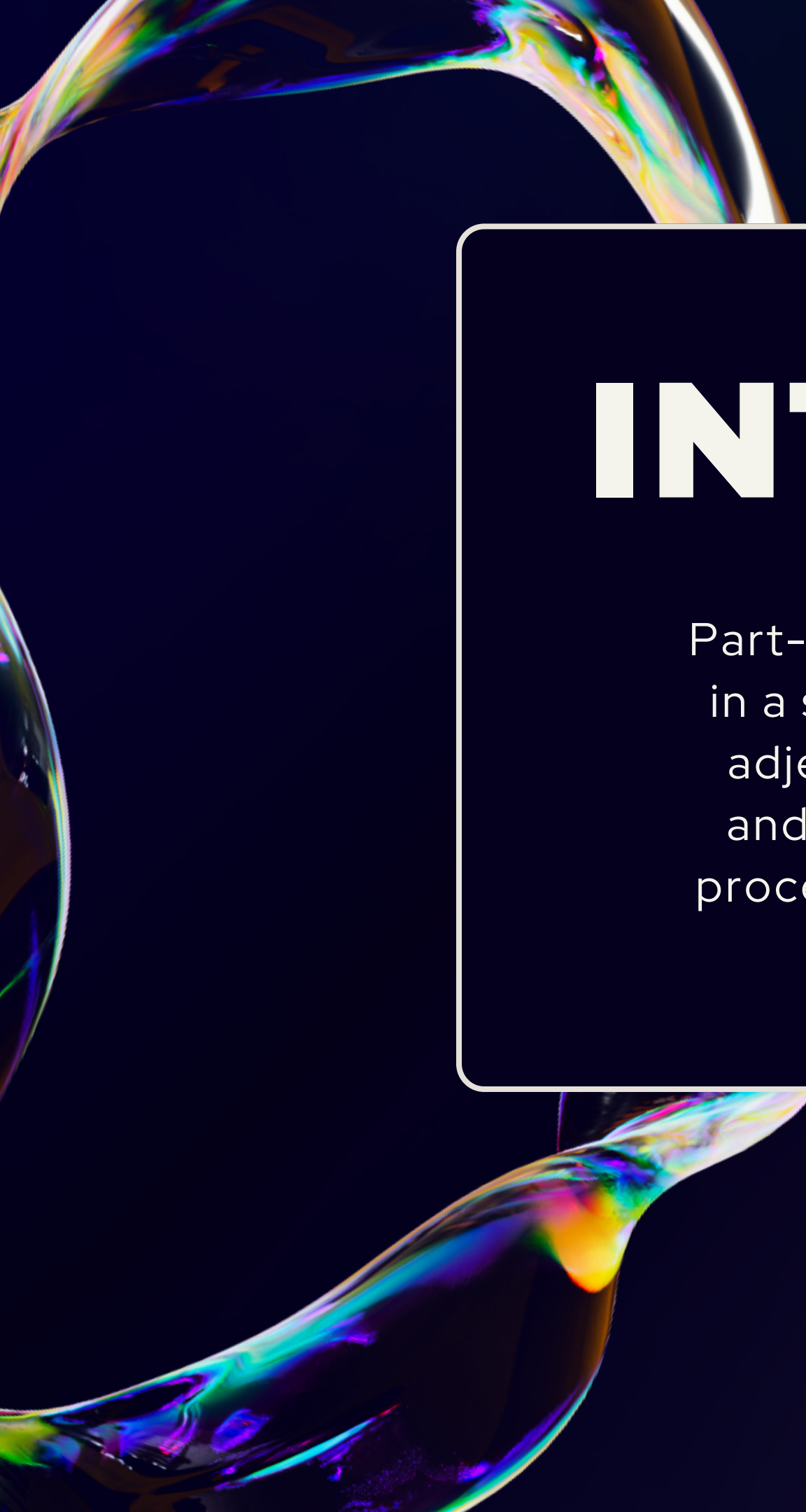
SEING Ratana

SIENG Lyheng

CHHEANG Udomveasna

HEANG Denis



An abstract, vibrant splash of liquid in shades of blue, green, yellow, and orange, appearing to flow from the top left and bottom left corners of the frame. The liquid has a glossy, reflective texture with internal ripples and highlights.

# INTRODUCTION

Part-of-speech tagging is the task of labeling every word in a sentence with its grammatical category: noun, verb, adjective, determiner, and so on. It's one of the oldest and most widely used components in natural language processing and machine translation because it's required to know each word's role.

# PROBLEM

## Traditional Approach

|      | N | V | M |
|------|---|---|---|
| Mary | 2 | 0 | 0 |
| see  | 0 | 3 | 0 |
| Jane | 2 | 0 | 0 |
| Will | 2 | 0 | 3 |

**Unigram**

|           | N-M | M-V | V-N |
|-----------|-----|-----|-----|
| mary-will | 1   | 0   | 0   |
| will-see  | 0   | 3   | 0   |
| see-jane  | 0   | 0   | 1   |
| will-will | 1   | 0   | 0   |
| see-mary  | 0   | 0   | 1   |
| jane-will | 1   | 0   | 0   |
| see-will  | 0   | 0   | 1   |

**Bigram**

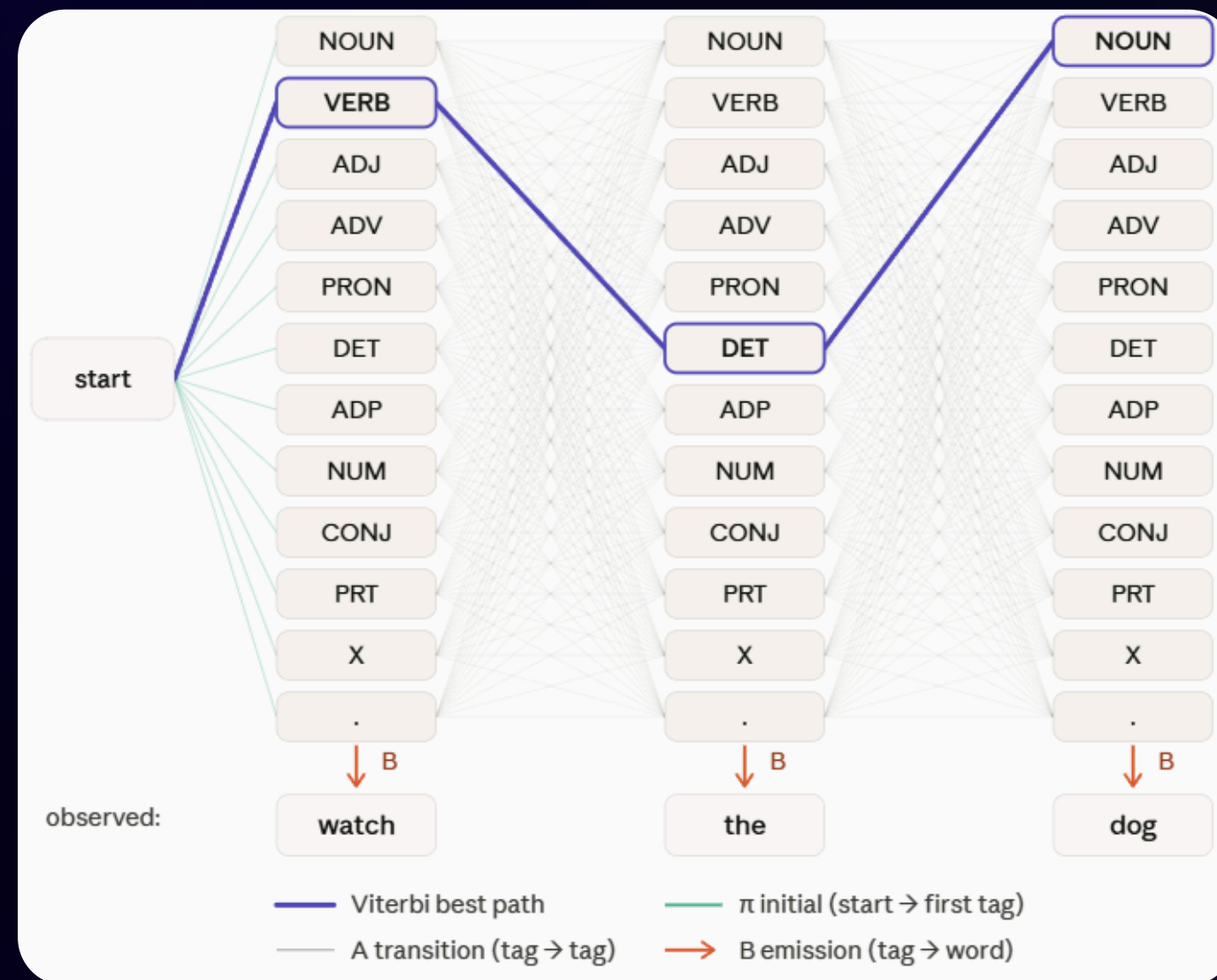
1. Use tables and rules. Unreliable and time consuming.

2. Fail when working with unseen words conflicting tags.

3. Doesn't learn and super complex to scale.

# SOLUTION

## HMM APPROACH



The idea is that an underlying sequence of tags "generates" the words you observe, and tagging means working backwards finding the hidden tag sequence most likely to have produced the words

A Hidden Markov Model treats a sentence as two parallel sequences.

- the words you can see (the observations)
- the grammatical tags you can't see (the hidden states).

The model is defined by three sets of probabilities.

- Initial ( $\pi$ ): how likely each tag is to start a sentence

$$\pi(s_i) = \frac{\text{Count}(\text{sentence begins with } s_i)}{M}$$

- Transition (A): how likely one tag is to follow another

$$P(s_j | s_i) = \frac{\text{Count}(s_i, s_j)}{\text{Count}(s_i)}$$

- Emission (B): how likely a given tag is to produce a given word

$$P_{\text{smooth}}(o_t | s_i) = \frac{\text{Count}(o_t, s_i) + \alpha}{\text{Count}(s_i) + \alpha (|V| + 1)}$$

The best tag is defined by joint probability.

$$Q^* = \arg \max_Q \pi(q_1) \prod_t P(q_t | q_{t-1}) P(o_t | q_t)$$

To find the maximum of joint probability earlier, we use Viterbi algorithm.

$$\log v_t(j) = \max_i \left[ \log v_{t-1}(i) + \log A(j | i) \right] + \log B(o_t | j)$$

# DATASET & METHODOLOGY

We use Universal POS tags:

- Total POS tags: 12

Example: ADJ, PRON, ., CONJ, NOUN, DET, ADP, VERB, ADV, X, NUM, PRT

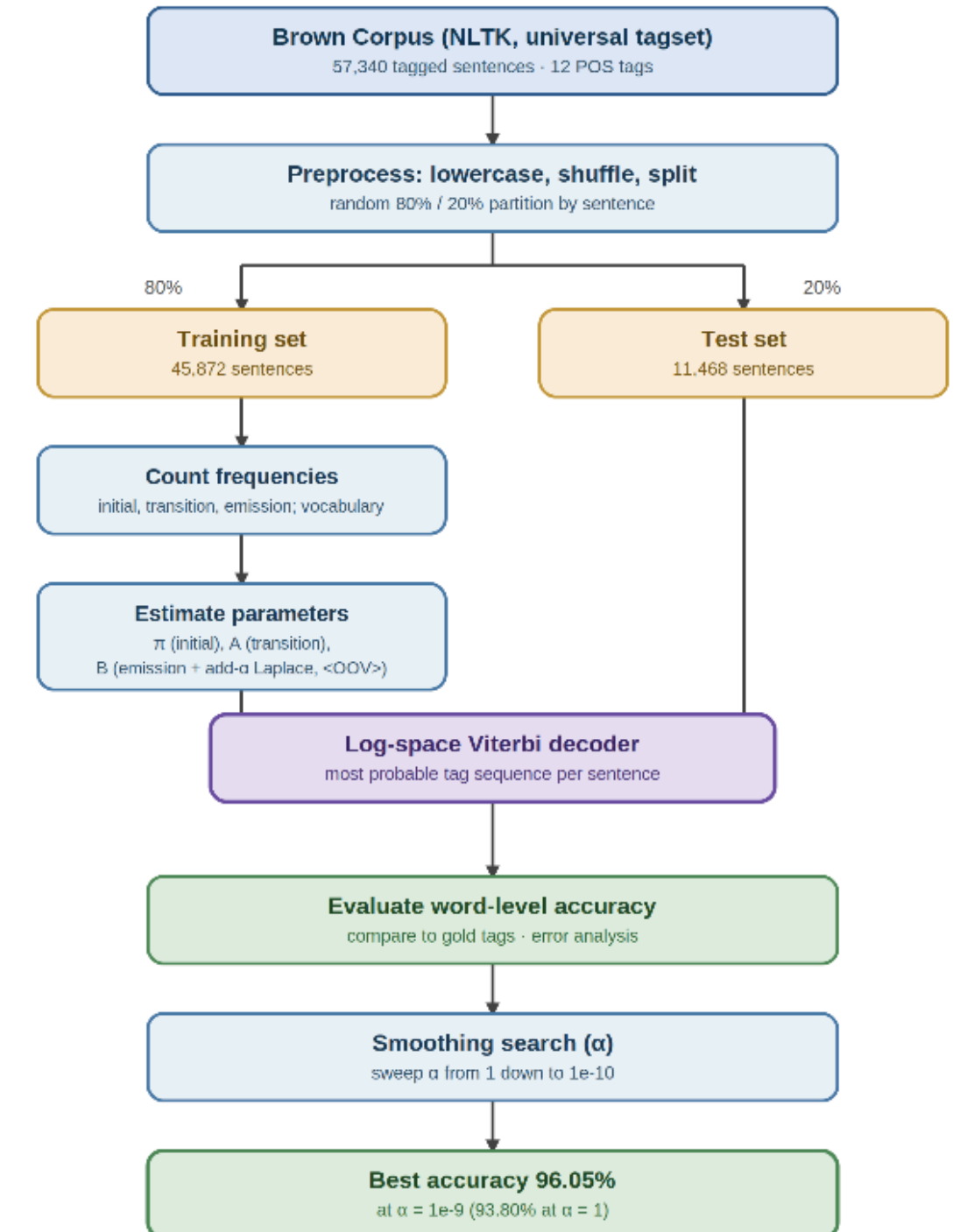
We use Universal POS tags:

- Total POS tags: 12

Example: ADJ, PRON, ., CONJ, NOUN, DET, ADP, VERB, ADV, X, NUM, PRT

During calculating Transition Probability (B), we performed grid search. We used 90% of training data for grid search training and another 10% for validation to find the best alpha

| Smoothing strength $\alpha$ | Validation accuracy |
|-----------------------------|---------------------|
| 1                           | 93.4885%            |
| $10^{-6}$                   | 95.4712%            |
| $10^{-7}$                   | 95.6834%            |
| $10^{-8}$                   | 95.8409%            |
| <b><math>10^{-9}</math></b> | <b>95.8529%</b>     |
| $10^{-10}$                  | 95.8507%            |
| $10^{-11}$                  | 95.8496%            |
| $10^{-12}$                  | 95.8496%            |



# RESULT

Table 1: Initial Probabilities ( $\pi$ )

|    | Tag  | P(tag at sentence start) |
|----|------|--------------------------|
| 1  | DET  | 0.213660                 |
| 2  | PRON | 0.159596                 |
| 3  | NOUN | 0.141502                 |
| 4  | ADP  | 0.123714                 |
| 5  | ADV  | 0.090426                 |
| 6  | .    | 0.087025                 |
| 7  | CONJ | 0.049289                 |
| 8  | VERB | 0.045823                 |
| 9  | PRT  | 0.037125                 |
| 10 | ADJ  | 0.034836                 |
| 11 | NUM  | 0.016437                 |
| 12 | X    | 0.000567                 |

Initial

Table 2: Transition Probabilities (A) [row = previous tag, col = current tag]

| From \ To | .      | ADJ    | ADP    | ADV    | CONJ   | DET    | NOUN   | NUM    | PRON   | PRT    | VERB   | X      |
|-----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| .         | 0.1721 | 0.0467 | 0.1057 | 0.0706 | 0.1122 | 0.1099 | 0.1332 | 0.0193 | 0.0747 | 0.0300 | 0.1235 | 0.0022 |
| ADJ       | 0.1002 | 0.0570 | 0.0883 | 0.0099 | 0.0382 | 0.0057 | 0.6525 | 0.0070 | 0.0039 | 0.0191 | 0.0176 | 0.0006 |
| ADP       | 0.0098 | 0.0832 | 0.0203 | 0.0156 | 0.0018 | 0.4556 | 0.2581 | 0.0301 | 0.0696 | 0.0143 | 0.0412 | 0.0005 |
| ADV       | 0.1700 | 0.1355 | 0.1425 | 0.0972 | 0.0175 | 0.0739 | 0.0324 | 0.0135 | 0.0478 | 0.0287 | 0.2408 | 0.0001 |
| CONJ      | 0.0208 | 0.1136 | 0.0720 | 0.0918 | 0.0003 | 0.1522 | 0.2407 | 0.0192 | 0.0673 | 0.0257 | 0.1957 | 0.0007 |
| DET       | 0.0129 | 0.2406 | 0.0090 | 0.0171 | 0.0007 | 0.0059 | 0.6260 | 0.0099 | 0.0101 | 0.0022 | 0.0641 | 0.0014 |
| NOUN      | 0.2846 | 0.0129 | 0.2447 | 0.0266 | 0.0599 | 0.0156 | 0.1505 | 0.0080 | 0.0198 | 0.0180 | 0.1592 | 0.0004 |
| NUM       | 0.2724 | 0.0603 | 0.1309 | 0.0207 | 0.0394 | 0.0135 | 0.3813 | 0.0217 | 0.0085 | 0.0052 | 0.0458 | 0.0003 |
| PRON      | 0.1031 | 0.0096 | 0.0567 | 0.0546 | 0.0113 | 0.0173 | 0.0090 | 0.0011 | 0.0081 | 0.0238 | 0.7054 | 0.0000 |
| PRT       | 0.0765 | 0.0192 | 0.0892 | 0.0359 | 0.0123 | 0.0848 | 0.0362 | 0.0053 | 0.0069 | 0.0111 | 0.6225 | 0.0001 |
| VERB      | 0.0807 | 0.0577 | 0.1692 | 0.1028 | 0.0146 | 0.1636 | 0.0976 | 0.0090 | 0.0547 | 0.0656 | 0.1843 | 0.0002 |
| X         | 0.2747 | 0.0034 | 0.0536 | 0.0077 | 0.0255 | 0.0060 | 0.0553 | 0.0009 | 0.0077 | 0.0043 | 0.0485 | 0.5128 |

Transition

Table 3: Emission Probabilities (B)

|    | Tag  | Most Probable Word | P(word   tag) |
|----|------|--------------------|---------------|
| 1  | .    | ,                  | 0.395997      |
| 2  | ADJ  | other              | 0.020364      |
| 3  | ADP  | of                 | 0.251409      |
| 4  | ADV  | not                | 0.082462      |
| 5  | CONJ | and                | 0.755699      |
| 6  | DET  | the                | 0.509387      |
| 7  | NOUN | time               | 0.005859      |
| 8  | NUM  | one                | 0.181696      |
| 9  | PRON | he                 | 0.194035      |
| 10 | PRT  | to                 | 0.502005      |
| 11 | VERB | is                 | 0.055550      |
| 12 | X    | de                 | 0.049278      |

Emission

# TESTING

| Configuration                                             | Tokens correct / total   | Word-level accuracy |
|-----------------------------------------------------------|--------------------------|---------------------|
| Add-one baseline ( $\alpha = 1$ )                         | 217,777 / 232,177        | 93.80%              |
| <b>Selected smoothing (<math>\alpha = 10^{-9}</math>)</b> | <b>223,015 / 232,177</b> | <b>96.05%</b>       |

After trained the HMM model using  $\alpha=1e-9$ , we test it on 11,468 sentences or 232,177 words tokens. The result shows that  $\alpha=1e-9$  achieved 2.25% higher accuracy than the recommended value ( $\alpha=1$ ).

# ERROR ANALYSIS

| Word                | Prediction | True | Likely cause                 |
|---------------------|------------|------|------------------------------|
| <i>non-partisan</i> | DET        | ADJ  | Unseen adjective (OOV)       |
| <i>chi-chi</i>      | .          | NOUN | Unseen noun (OOV)            |
| <i>offer</i>        | VERB       | NOUN | Noun–verb ambiguity          |
| <i>tempting</i>     | NOUN       | VERB | Participle read as a noun    |
| <i>very</i>         | ADJ        | ADV  | Adverb / adjective ambiguity |
| <i>as</i>           | ADV        | ADP  | Function-word ambiguity      |



# **LIMITATION**

## **The first-order Markov assumption**

The tagger is a bigram (first-order) HMM, can't model dependencies beyond the immediately preceding tag.

## **The output-independence assumption**

The model assumes a word depends only on its own tag, and treats every word as an atomic, indivisible symbol.

- hyphen (non-partisan)
- capital letter (proper nouns)
- characteristic suffix (-ing, -ly, -tion)

## **Out-of-vocabulary words**

Every unseen word gets the exact same emission profile. To your model they are all just "`<OOV>`". The emission table literally cannot tell them apart.

# CONCLUSION

This work built an HMM POS tagger from scratch on the Brown Corpus. It reached 93.80% accuracy, rising to 96.05% after tuning  $\alpha$ . The remaining errors come from two intrinsic limits: unseen words and dependencies beyond the first-order Markov assumption.

